

PAPER_850: ADD Extra Dimensions Extended — Graviton Leakage and Negative Buoyancy at Sgr A*

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Abstract

The Arkani-Hamed-Dimopoulos-Dvali (ADD) large extra dimensions model (arXiv:1607.01831) is extended within UQFF via the F_{LED} term. With $n=2$ extra dimensions and $M^* = 1$ TeV, $F_{LED} = 6.72e-23$ N — negligible against F_{LENR} ($6.17e37$ N) but conceptually significant at negative-buoyancy SMBH sites. The graviton leakage hypothesis proposes that extra-dimensional drainage amplifies vacuum reversal at Sgr A*, connecting sub-millimeter gravity tests ($R \sim 0.2$ mm compactification radius) to astrophysical negative buoyancy.

1. ADD Model Framework

ADD (1998): gravity propagates in $(4+n)$ dimensions
SM fields confined to $3+1$ brane

Fundamental scale: $M^* \sim 1$ TeV (accessible to LHC)
Planck scale: $M_{Pl} = 1.22e19$ GeV (apparent 4D weakness of gravity)

Relationship: $M_{Pl}^2 \sim M^{*(n+2)} * R^n$
For $n=2$: $R \sim (M_{Pl}/M^*)^{(2/2)} / M^* * \hbar c$
 $= (1.22e19 / 10^3)^1 / 10^3 * 1.973e-16$
 $\sim 2.41e-4$ m ~ 0.24 mm

2. F_{LED} Term

$F_{LED} = k_{LED} * (M^* / M_{Pl})^2$
 $= 10^{10} * (10^3 / 1.22e19)^2$
 $= 10^{10} * 6.72e-34$
 $= 6.72e-24$ N

Comparison:
 $F_{LED} = 6.72e-24$ N
 $F_{LENR} = 6.17e37$ N
 $F_{LED}/F_{LENR} = 1.09e-61$

3. Compactification Radius

$R_n \sim (M_{Pl} / M^*)^{(2/n)} * \hbar c / M^*$

$n=1$: $R \sim 2.41e13$ m (too large, ruled out by solar system)
 $n=2$: $R \sim 1.96e-4$ m (0.2 mm — submillimeter gravity test range)
 $n=3$: $R \sim 6.8e-12$ m (atomic scale)

n=4: $R \sim 1.6e-15$ m (nuclear scale)
n=6: $R \sim 1.1e-19$ m (electroweak scale)

n=2 is the phenomenologically interesting case:
submillimeter gravity tests (Eot-Wash, IUPUI) constrain $R < \sim 37$ μ m
for n=2, pushing $M^* > 3.6$ TeV.

4. Graviton Leakage at Sgr A*

Hypothesis:

At Sgr A* (negative buoyancy, $F_{U_Bi} \sim -8.31e211$ N),
graviton leakage into n=2 extra dimensions may amplify
the vacuum reversal mechanism.

$F_{U_Bi}(\text{Sgr A}^*, \text{ with } F_{LED}) = F_{U_Bi}(\text{base}) + F_{LED}$
 $= -8.31e211 + 6.72e-24$
 $\sim -8.31e211$ N (F_{LED} negligible numerically)

But: the ADD framework provides a theoretical mechanism
for WHY negative buoyancy occurs at SMBH scales:
graviton drainage reduces effective 4D gravitational coupling
at short distances, while UQFF's F_0 scale reversal amplifies
the effect at astrophysical scales.

5. 8-System Recalculation with F_{LED}

All 8 sonification systems recalculated with F_{LED} integrated:

$F_{LED} = 6.72e-24$ N (constant across all systems)

Impact: $<10^{-60}$ fractional change in F_{U_Bi} for all systems
Physically interesting but numerically negligible.

Conclusion

F_{LED} connects UQFF to the ADD extra-dimensional framework at $6.72e-24$ N. While numerically negligible, the graviton leakage hypothesis at Sgr A* provides a theoretical bridge between submillimeter gravity experiments and astrophysical negative buoyancy. The n=2 compactification radius of ~ 0.2 mm sits precisely in the range of current torsion balance experiments.

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